its potential energy bowl has the right shape for inflation to have occurred, why there are space and time within which the whole discussion takes place, and, in Leibniz's more grandiose phrasing, why there is something rather than nothing.

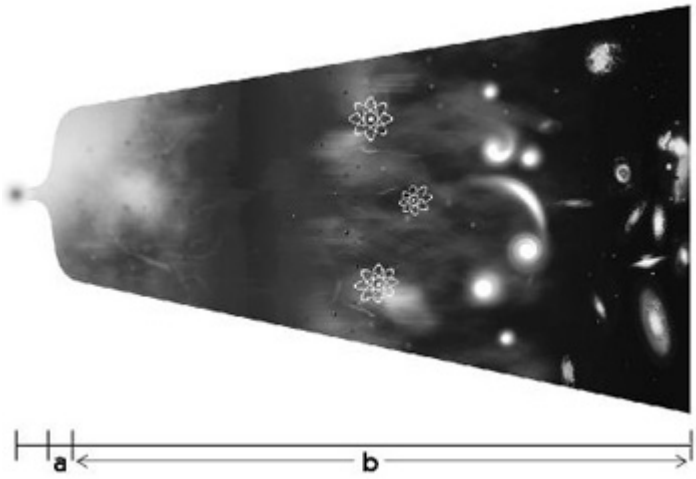


Figure 10.3 **(**a**)** Inflationary cosmology inserts a quick, enormous burst of spatial expansion early on in the history of the universe. **(** b**)** After the burst, the evolution of the universe merges into the standard evolution theorized in the big bang model.

A second and related observation is that inflationary cosmology is not a single, unique theory. Rather, it is a cosmological framework built around the realization that gravity can be repulsive and can thus drive a swelling of space. The precise details of the outward burst—when it happened, how long it lasted, the strength of the outward push, the factor by which the universe expanded during the burst, the amount of energy the inflaton deposited in ordinary matter as the burst drew to a close, and so on—depend on details, most notably the size and shape of the inflaton field's potential energy, that are presently beyond our ability to determine from theoretical considerations alone. So for many years physicists have studied all sorts of possibilities—various shapes for the potential energy, various numbers of inflaton fields that work in tandem, and so on—and determined which choices give rise to theories consistent with astronomical observations. The important thing is that there are aspects of inflationary cosmological theories that transcend the details and hence are common to essentially any realization. The outward burst itself, by definition, is one such feature, and hence any inflationary model comes